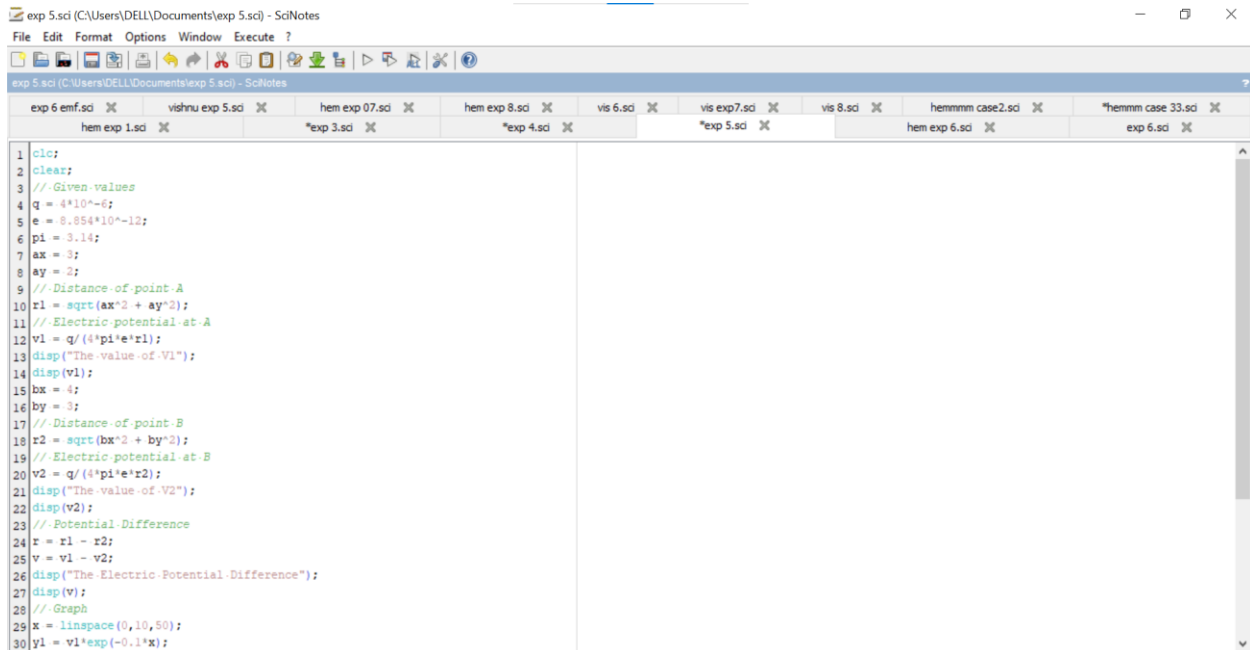
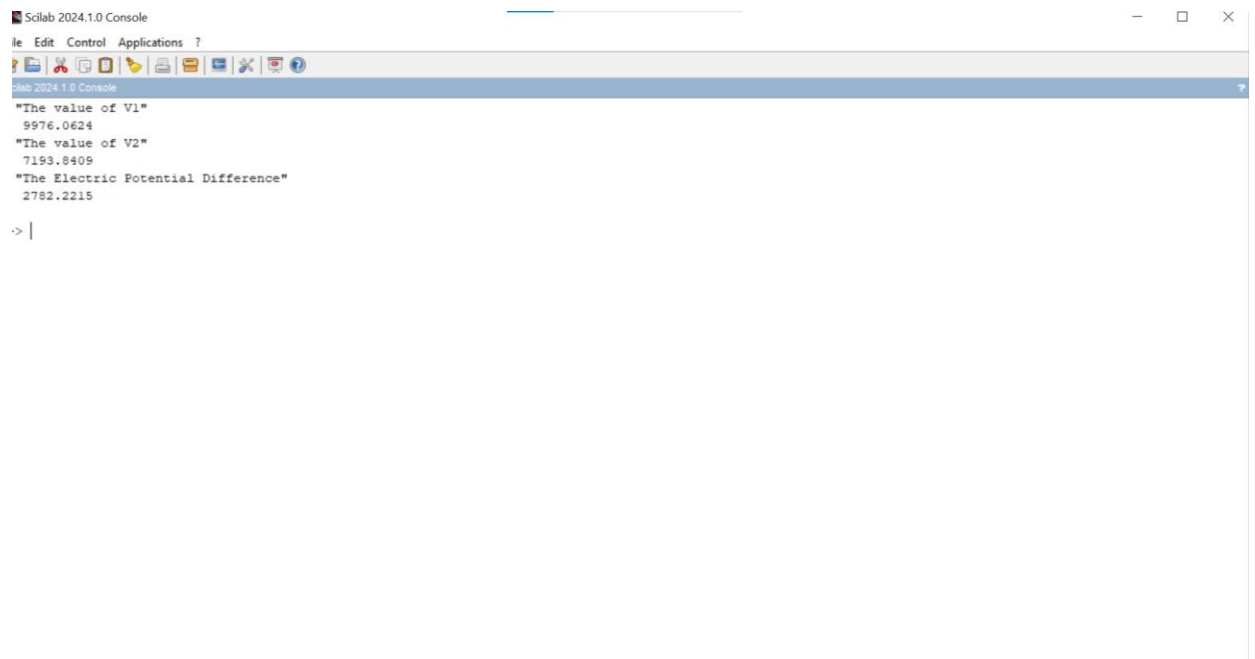


EXP 5: Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.



```
exp 5.sci (C:\Users\DELL\Documents\exp 5.sci) - SciNotes
File Edit Format Options Window Execute ?
exp 5.sci (C:\Users\DELL\Documents\exp 5.sci) - SciNotes
exp 6 emf.sci x vishnu exp 5.sci x hem exp 07.sci x hem exp 8.sci x vis 6.sci x vis exp7.sci x vis 8.sci x hemmm case2.sci x *hemmm case 33.sci x
hem exp 1.sci x *exp 3.sci x *exp 4.sci x *exp 5.sci x hem exp 6.sci x exp 6.sci x
1 clc;
2 clear;
3 // Given values
4 q = 4*10^-6;
5 e = 8.854*10^-12;
6 pi = 3.14;
7 ax = 3;
8 ay = 2;
9 // Distance of point A
10 r1 = sqrt(ax^2 + ay^2);
11 // Electric potential at A
12 v1 = q/(4*pi*e*r1);
13 disp("The value of V1");
14 disp(v1);
15 bx = 4;
16 by = 3;
17 // Distance of point B
18 r2 = sqrt(bx^2 + by^2);
19 // Electric potential at B
20 v2 = q/(4*pi*e*r2);
21 disp("The value of V2");
22 disp(v2);
23 // Potential Difference
24 r = r1 - r2;
25 v = v1 - v2;
26 disp("The Electric Potential Difference");
27 disp(v);
28 // Graph
29 x = linspace(0,10,50);
30 y1 = v1*exp(-0.1*x);
```



```
Scilab 2024.1.0 Console
File Edit Control Applications ?
Scilab 2024.1.0 Console
"The value of V1"
9976.0624
"The value of V2"
7193.8409
"The Electric Potential Difference"
2782.2215
-> |
```

Ex 5:-

$Q = 4 \times 10^{-6}$

$a_x = 3$

$a_y = 2$

$b_x = 4$

$b_y = 3$

Sol:

$$r_1 = \sqrt{a_x^2 + a_y^2} = b$$

$$r_1 = \sqrt{3^2 + 2^2}$$

$$r_1 = \sqrt{9+4}$$

$$r_1 = \sqrt{13} = 3.6055 \text{ m}$$

$$V_1 = 4 \times 10^{-6}$$

$$4(3.14)(8.854 \times 10^{-12})(3.6055)$$

$$V_1 \approx 9977.9 \text{ V}$$

$$r_2 = \sqrt{b_x^2 + b_y^2}$$

$$r_2 = \sqrt{4^2 + 3^2}$$

$$r_2 = \sqrt{16+9}$$
$$r_2 = 5 \text{ m}$$

$$V_2 = \frac{q}{4\pi\epsilon_0 r^2}$$

$$V_2 = \frac{4 \times 10^{-6}}{4(3.14)(8.854 \times 10^{-12})(5)}$$

$$V_2 \approx 7191.3 \text{ V}$$

$$\Delta V = V_1 - V_2$$

$$\Delta V = 9977.9 - 7191.3$$

$$\Delta V \approx 2786.6 \text{ V}$$

O/P ✓